CHAPTER 4

# Differentiation: from rate to tangent

**Key point · Chapter path. Objectives.** Interpret a rate; approximate a derivative at a point; determine a tangent; understand why differentiability fails.

**Prerequisites.** Function values, slope of a line, algebraic manipulation, square root and absolute value.

**Yearly reference point.** 3–12 November: 8 hours

## Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. What is the slope of the line through $(1,2)$ and $(3,8)$?
2. Expand $(2+h)^2-4$.
3. Simplify $\dfrac{4h+h^2}{h}$ if $h \neq 0$.
4. Calculate $\sqrt{0}$ and $\sqrt{0.01}$.
5. Calculate $|-3|$ and $|2|$.
6. A line of slope 2 passes through $(1,5)$. What is its equation?

**After the diagnostic.** Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

**Speaking and logic.** Explain to a classmate why we simplify the rate before letting h tend to 0. Distinguish average velocity from instantaneous velocity.

## Investigate before formalising

For $f(x) = x^2$, let h vary, positive and negative, around 0, keeping the point A with x-coordinate 1. Draw three secants; conjecture a limiting line, then determine it without reading a graph.

**Organisation.** Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.

## 4.1 Rate of change and secants

**Key point · Definition and results.** For $a \neq b$ in the domain of f, the rate of change between a and b is $\dfrac{f(b)-f(a)}{b-a}$. It is the slope of the secant through the corresponding points. In context, the unit of the rate is the unit of f divided by that of the variable.

**Status and justification.** Definition; formula for the slope of a line, already studied. With $b = a+h$, $h \neq 0$, the rate becomes $\dfrac{[f(a+h)-f(a)]}{h}$.

**Example 1.** For $f(x) = x^2$, between 1 and 3, the rate is $\dfrac{9-1}{3-1} = 4$. Between 1 and $1+h$, it is $2+h$. The slopes for $h = 0.1$ and $h = -0.1$ are 2.1 and 1.9.

**Watch out.** The rate is not $\dfrac{f(b)}{f(a)}$; h represents a gap, positive or negative depending on the domain.

**Application.** Calculate the rate of $f(x) = x^2 + 3x$ between 2 and $2+h$.

## 4.2 Derivative at a point

**Key point · Definition and results.** f is differentiable at a when the rate $\dfrac{[f(a+h)-f(a)]}{h}$ tends to a real number as h tends to 0, within the domain. This real number is $f\prime(a)$. An intuitive approach to the limit is used here. In kinematics, if f gives a position in metres and time is in seconds, the derivative at a point represents an instantaneous velocity in $\mathrm{m\,s^{-1}}$.

**Status and justification.** For $f(x) = x^2$, the rate at a is $2a+h$, which gets closer to $2a$; hence $f\prime(a) = 2a$. This calculation will be taken up again as a derivative function in chapter 6.

**Example 2.** For $s(t) = t^2 + 3t$, the rate around 2 is $7+h$, so $s\prime(2) = 7\ \mathrm{m\,s^{-1}}$. The average velocity on $[2,3]$ is $8\ \mathrm{m\,s^{-1}}$; it is not the velocity at time 2.

**Watch out.** We do not replace h by 0 in an unsimplified quotient: we would get a division by 0.

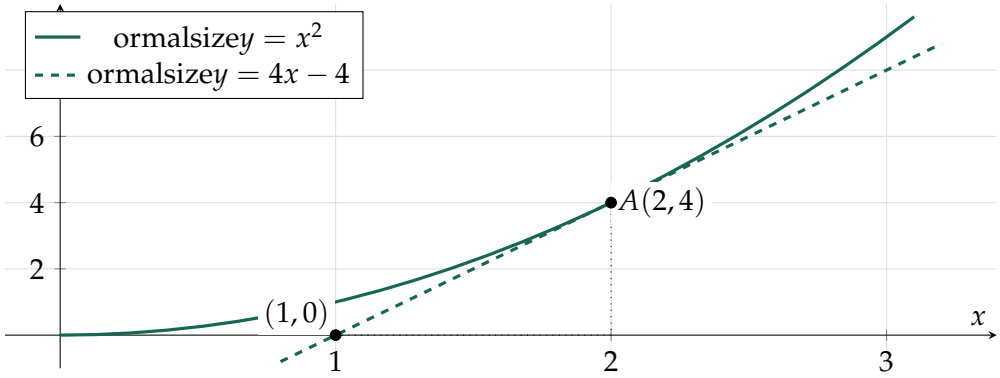
**Application.** For $f(x) = 3x^2 - 2x$, calculate $f\prime(1)$ from the rate.

## 4.3 Tangent and affine approximation

**Key point · Definition and results.** If f is differentiable at a, the tangent at the point $(a, f(a))$ has equation $y = f(a) + f\prime(a)(x-a)$. For small h, we use $f(a+h) \approx f(a) + f\prime(a)h$. This local approximation does not by itself provide a bound on the error.

**Status and justification.** The tangent has slope $f\prime(a)$ and passes through $(a, f(a))$. In $y = f\prime(a)x + p$, substituting this point gives $p = f(a) - af\prime(a)$, hence the equation.

**Example 3.** For $f(x) = x^2$ at 2, tangent $y = 4 + 4(x-2) = 4x - 4$. We approximate $2.01^2$ by $4 + 4(0.01) = 4.04$; the exact value is 4.0401, error $0.0001 = h^2$.




The tangent at A also passes through $(1,0)$. Its slope is $\dfrac{4-0}{2-1} = 4$. The dashed line distinguishes it from the curve, even in black and white.

**Watch out.** A tangent may cross the curve again elsewhere. "Only one common point" is not a definition.

**Application.** Given $f(3) = 5$ and $f\prime(3) = -2$. Write the tangent and approximate $f(3.04)$.

## 4.4 When differentiability fails

**Key point · Definition and results.** The square root function is defined on $[0, +\infty)$ but is not differentiable at 0. The absolute value function is defined on $\mathbb{R}$ but is not differentiable at 0. Being defined, or having a continuous curve, therefore does not guarantee differentiability.

**Status and justification.** For $h > 0$, $\dfrac{\sqrt{h} - 0}{h} = \dfrac{1}{\sqrt{h}}$, which does not tend to a finite real number as h gets closer to 0. For $|x|$, the rate $\dfrac{|h|}{h}$ equals 1 on the right and $-1$ on the left: no common value.

**Example 4.** For $|x|$, the secants from the origin all have slope 1 on the right and $-1$ on the left. The corner point does not have a unique tangent with finite slope. For the square root, the limiting direction is vertical; there is no real derivative.

**Watch out.** A very large slope is not a proof of non-differentiability; the behaviour of the rate must be explained.

**Application.** Justify that $g(x) = |x - 2|$ is not differentiable at 2. Also study the differentiability of $x \mapsto \sqrt{x}$ at 0 through its rate.

## Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

### Exercise 4.1 · Two secants

The function $f$ is defined on $\mathbb{R}$ by $f(x) = x^2 - 1$. Let $A$ be the point of its curve with x-coordinate 1.

1. Calculate the slope of the secant joining $A$ to the point of the curve with x-coordinate 2.
2. Repeat the calculation with the point of the curve with x-coordinate 1.1.
3. After studying the derivative at a point, determine $f'(1)$ and compare this value with the two rates calculated. Explain what changes when the second point gets closer to $A$.

### Exercise 4.2 · A tangent

Consider the function $f(x) = 2x^2 + 1$, defined on $\mathbb{R}$. The aim is to determine the equation of its tangent at the point with x-coordinate $-1$, starting from the rate of change.

1. For $h \neq 0$, write then simplify $\dfrac{f(-1+h) - f(-1)}{h}$.
2. Deduce $f'(-1)$. Also calculate $f(-1)$.
3. Write the slope-intercept equation of the tangent and check that it passes through the point of contact.

### Exercise 4.3 · Marginal cost

A company uses the model $C(q) = q^2 + 20q + 100$, where $q \geq 0$ is the quantity produced and $C(q)$ its cost in monetary units. The model extends the quantity to a real variable so as to study small changes in production.

1. Calculate $C(10)$ and $C(11)$, then the rate of change of $C$ between 10 and 11. Interpret this rate in context.
2. From the rate between 10 and $10 + h$, determine $C'(10)$.
3. Compare the two results. Explain why the cost of one extra unit is not exactly equal to the derivative in this model.

### Exercise 4.4 · Reading a tangent

This exercise has two independent situations. A tangent to a curve is a line: another point of this line does not necessarily belong to the curve.

1. On the figure in the section on the tangent, read the coordinates of two points of the green line. Deduce its slope and check the value stated in the legend.
2. In another situation, the tangent to the curve of a function $f$ at the point $A(2,3)$ passes through $B(4,2)$. Determine $f(2)$ and $f'(2)$. Can we deduce $f(4)$?
3. Use this tangent to suggest an approximate value of $f(2.1)$. Make the "approximately equal" sign appear in the conclusion.

### Exercise 4.5 · Checking an approximation

We wish to approximate $0.98^2$ without first carrying out the multiplication. We use the tangent to the curve of the square function at the point with x-coordinate 1.

1. Write the equation of this tangent and deduce the required approximation.
2. Then calculate the exact value of $0.98^2$. Define the error as "exact value minus approximate value": calculate this error and state its sign.
3. Explain the result using the identity $(1+h)^2 = 1 + 2h + h^2$.

### Exercise 4.6 · Tangent and intersection

The function $f$ is defined on $\mathbb{R}$ by $f(x) = x^3$. We accept that $f'(1) = 3$. A student claims that a tangent and a curve always have a single common point.

1. Determine the equation of the tangent to the curve of $f$ at the point with x-coordinate 1.
2. Check by expanding that $x^3 - 3x + 2 = (x-1)^2(x+2)$.
3. Determine all the points of intersection of the curve and this tangent. Is the student's claim correct?

### Exercise 4.7 · Numerical secants

To study the square function near 1, define, for $h \neq 0$,

$$T(h) = \frac{(1+h)^2 - 1}{h}.$$

We will use the six values 0.1, 0.01, 0.001 and their opposites.

1. Write a Python function that calculates $T(h)$, then produce a table of six results.
2. Conjecture the value of the derivative of the square function at 1.
3. Simplify $T(h)$ algebraically to justify the conjecture. Explain why trying numerical values is not, on its own, enough to prove the result.

### Exercise 4.8 · Delayed review

The reciprocal function is defined by $f(x) = \dfrac{1}{x}$ for $x \neq 0$. We study it near 1.

1. State the forbidden values of $h$ in the rate $\dfrac{f(1+h) - f(1)}{h}$, then simplify it.
2. Deduce $f\prime(1)$ and the equation of the tangent at 1.
3. Use this tangent to approximate $\dfrac{1}{1.02}$. Keep the distinction between exact value and approximate value.

**Homework.** Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

## To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

**CQFD, exercise 17 p. 127, all questions.** Read two points on each tangent; write the quotient of the increments rather than estimating its steepness.

**CQFD, exercise 19 p. 127, questions 1 and 2.** Distinguish the y-coordinate of the point of contact from the slope of the tangent line.

## MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. Rate of $x^2$ between 1 and 2:
   A. 2
   B. 3
   C. 1
   D. 4
2. If $f(1) = 4$, $f'(1) = 2$, the tangent is:
   A. $y = 2x + 2$
   B. $y = 4x + 2$
   C. $y = 2x + 4$
   D. $y = x + 3$
3. The derivative at a point is
   A. a function
   B. an area
   C. an x-coordinate
   D. a real number if f is differentiable at a
4. For $x^2$ at 3, the derivative at that point is:
   A. 9
   B. 3
   C. 6
   D. 12
5. The approximation of $f(a + h)$ is
   A. $f(a)f'(a)$
   B. $f(a) + hf'(a)$
   C. $f(a) + f'(h)$
   D. $hf(a)$

## MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $\sqrt{x}$ at 0 is
   A. differentiable with derivative 0
   B. differentiable with derivative 1
   C. not differentiable
   D. not defined
2. $\dfrac{|h|}{h}$ for $h < 0$ equals
   A. $-1$
   B. 1

C. $0$
D. $h$

3. A tangent of slope $0$ is
   A. vertical
   B. oblique
   C. impossible
   D. horizontal
4. Position in m and time in s: the unit of the derivative is
   A. $\mathrm{m}^2$
   B. $\mathrm{m\,s}^{-1}$
   C. $\mathrm{s\,m}^{-1}$
   D. m
5. For f differentiable at a, replacing $f(a+h)$ by $f(a)+hf\prime(a)$ is in general
   A. an exact identity
   B. a definition of h
   C. a local approximation
   D. always false, even for an affine function

## True or false? Justify.

1. A tangent can never cross a curve again.
2. The absolute value function is differentiable at $0$.
3. An affine function $mx+p$ has derivative m at every real number.

**Key point · Self-assessment.** I give the domain and h nonzero; I read a slope; I use the point of tangency; I distinguish exact from approximate.

For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.

# Solutions and teaching notes for chapter 4

**Teacher use.** The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

## Diagnostic and remediation

1. $\dfrac{8-2}{3-1} = 3$. *Action:* Represent the horizontal and vertical increments.
2. $4h + h^2$. *Action:* Expand before simplifying.
3. $4 + h$. *Action:* Factor h out of the numerator before dividing.
4. 0 and 0.1. *Action:* Check by squaring.
5. 3 and 2. *Action:* Represent the distance to 0.
6. $y = 2x + 3$. *Action:* Substitute the coordinates to find the y-intercept.

## Investigation activity

The slopes are $2 + h$; limit 2. The line passes through $(1, 1)$, so $y = 2x - 1$. The drawing motivates, the calculation justifies.

## Immediate applications

1. $f(2 + h) - f(2) = 7h + h^2$; rate $7 + h$ for $h \neq 0$.
2. The rate is $4 + 3h$, so the derivative at 1 is 4.
3. $y = 5 - 2(x - 3) = -2x + 11$; approximation $5 - 2(0.04) = 4.92$.
4. The rate at 2 is $\dfrac{|h|}{h}$, with two opposite limiting values. For the square root at 0, the rate $\dfrac{1}{\sqrt{h}}$ for $h > 0$ exceeds every bound as h gets closer to 0: no real derivative.

## Exercises: method and results

**Exercise 4.1 – Two secants.** Slopes 3 and 2.1; derivative at 1 is 2. The second is closer to the slope of the tangent; this is an exact calculation of a rate.

**Exercise 4.2 – A tangent.** Rate $4a + 2h = -4 + 2h$, so $f\prime(-1) = -4$. $f(-1) = 3$; tangent $y = 3 - 4(x + 1) = -4x - 1$.

**Exercise 4.3 – Marginal cost.** $C(10) = 400$, $C(11) = 441$, increase 41 for one unit. The rate around 10 is $40 + h$, so derivative 40: an approximation of the increase for a small change, not an equality for a whole unit.

**Exercise 4.4 – Reading a tangent.** On the figure in the lesson: points (1;0) and (2;4), slope 4. In the new situation, $f(2) = 3$, slope $-\dfrac{1}{2}$. Approximation $3 - 0.5 \times 0.1 = 2.95$. The y-coordinate of B is not necessarily $f(4)$.

**Exercise 4.5 – Checking an approximation.** $1 + 2(-0.02) = 0.96$. Exact 0.9604; exact minus approximate $0.0004 > 0$. The omitted term is $h^2$: approximation from below.

**Exercise 4.6 – Tangent and intersection.** The tangent has equation $y = 1 + 3(x-1) = 3x - 2$. Expanding the product indeed gives $x^3 - 3x + 2$. The x-coordinates of the intersections are therefore 1 and $-2$, by the zero-product rule. The points are $(1,1)$ and $(-2,-8)$. There are two common points: that is enough to refute the general claim.

**Exercise 4.7 – Numerical secants.** List of rates $2 + h$: 2.1; 2.01; 2.001 and 1.9; 1.99; 1.999. The algebraic calculation gives the real number 2. For very small h, floating-point calculation can lose precision.

**Exercise 4.8 – Delayed review.** For $h \neq 0, -1$, the rate is $-\dfrac{1}{1+h}$; it tends to $-1$. Approximation $1 - 0.02 = 0.98$. The exact value $\dfrac{50}{51}$ allows a check.

**CQFD, exercise 17 p. 127, all questions.** The x-coordinates of A, B and C are 0, 1 and 2. The tangents give slopes 2, 0 and $-2$ respectively. Thus $f'(0) = 2$, $f'(1) = 0$, $f'(2) = -2$. The units of the axes must be read from the grid.

**CQFD, exercise 19 p. 127, questions 1 and 2.** The point A reads $(1,4)$, so $f(1) = 4$. The tangent passes through $(0,1)$ and $(1,4)$: slope 3. Thus $f'(1) = 3$ and the tangent has equation $y = 4 + 3(x-1) = 3x + 1$.

## MCQ: answers and reasons

**MCQ 1.**

1. B. $\dfrac{4-1}{2-1} = 3$.
2. A. $4 + 2(x-1)$.
3. D. It is a value at a point.
4. C. Rate $6 + h$.
5. B. Tangent affine function.

**MCQ 2.**

1. C. Rate $\dfrac{1}{\sqrt{h}}$ not finite.
2. A. $|h| = -h$.
3. D. Constant y-coordinate.
4. B. A change in position divided by a change in time is expressed in $\mathrm{m\,s^{-1}}$.
5. C. Exact equality for an affine function, approximation in general.

### True or false

1. False: $x^3$ and the tangent $3x - 2$ at 1 also meet at $-2$.
2. False: the rates on either side are $-1$ and 1.
3. True: the rate is exactly m for every nonzero h.

**Differentiated teaching.** For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.